

Step-by-Step Practical Procedure Guide

Automatic Night Light System (LDR & Arduino UNO R3)

Department of Computer Science & Engineering / IoT Laboratory

Course Code: PCC-CS791 • Academic Session: 2025–2026 • Degree: B.Tech

Overview & Purpose of this Guide

This guide provides an easy-to-follow technical walkthrough to request components with official commercial part numbers, estimate hardware costs in Indian Rupees (INR / Rs.), assemble the circuit using a clear graphical wiring diagram, flash the firmware, and calibrate the system.

Phase 1: Component Requisition, Part Numbers & Costs (INR)

Present this official **Bill of Materials (BOM) & Cost Table** to your professor or lab store in-charge to requisition the components:

#	Component Full Name	Part ID / Code	Technical Ratings	Qty	Cost (Rs.)
1	Arduino UNO R3 Board	A000066	ATmega328P MCU, 16 MHz, 5V, 10-bit ADC	1	Rs. 480.00
2	CdS Photoresistor (LDR)	GL5528	5 mm Cell, $R_{10} = 10 - 20 \text{ k}\Omega$, $R_{\text{dark}} \geq 1 \text{ M}\Omega$	1	Rs. 10.00
3	Resistor (Pull-Down)	CFR-25JR-10K	$10 \text{ k}\Omega \pm 5\%$, 0.25 W (Brown-Black-Orange)	1	Rs. 1.50
4	Resistor (LED Limiter)	CFR-25JR-220R	$220 \Omega \pm 5\%$, 0.25 W (Red-Red-Brown-Gold)	1	Rs. 1.50
5	5mm Diffused LED	WP7113ID	$V_F \approx 2.0 \text{ V}$, $I_F = 20 \text{ mA}$, Red	1	Rs. 3.00
6	Solderless Breadboard	MB-102	830 Tie-Points (630 Terminal + 200 Bus)	1	Rs. 110.00
7	Jumper Wires (M-to-M)	JW-MM-20CM	24 AWG, 20 cm length, 2.54 mm Dupont	6	Rs. 20.00
8	USB Programming Cable	CAB-USB-AB-1M	USB Type-A to Type-B Shielded Cable, 1 m	1	Rs. 50.00
Total Estimated Hardware Cost:					Rs. 676.00

Table 1: Technical Requisition Matrix with Commercial Part IDs and Market Costs (INR).

Step 1.1: Component Terminal & Polarity Identification

- **Arduino UNO R3 (A000066):** Locate pin headers **5V**, **GND**, Analog In **A0**, and Digital Out **D9**.
- **LDR Photoresistor (GL5528):** Symmetrical 2-lead optoelectronic device (**no polarity**); either leg connects to +5 V or A0.
- **10 k Ω Divider Resistor (CFR-25JR-10K):** Bands: **Brown (1) – Black (0) – Orange ($\times 10^3$) – Gold ($\pm 5\%$)**.
- **220 Ω LED Resistor (CFR-25JR-220R):** Bands: **Red (2) – Red (2) – Brown ($\times 10^1$) – Gold ($\pm 5\%$)**.
- **5mm LED Indicator (WP7113ID):** **Longer Lead (Anode +)** connects to D9 via 220 Ω ; **Shorter Lead (Cathode –)** connects to **GND**.

Phase 2: Graphical Wire Connection Diagram

Below is the complete graphical wiring diagram showing how the **Arduino UNO R3** connects to the **Solderless Breadboard**. All four interconnect jumper wires are color-coded and routed with clear, non-intersecting trajectories for direct visual tracing:

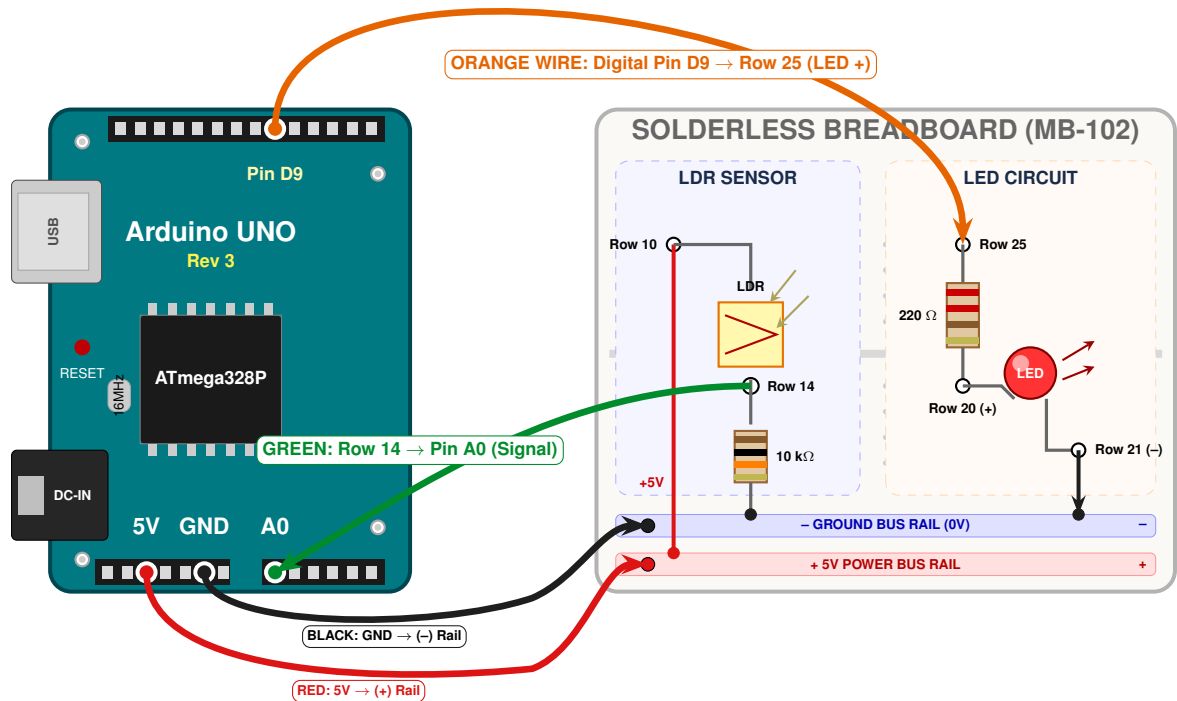


Figure 1: Graphical Wire Connection Diagram showing Arduino UNO R3 Header-to-Breadboard Connections.

Wire Color	Origin Pin (Arduino)	Destination Point (Breadboard)	Circuit Function / Signal
Orange	Pin D9 (Digital PWM Header)	Row 25 (Top Lead of 220 Ω Resistor)	Switched Output Gate Drive (+5V Active)
Green	Pin A0 (Analog In Header)	Row 14 (Divider Tap Junction Node)	Analog Light Sensor Voltage Signal
Black	Pin GND (Power Header)	Blue Bus Rail (-) (Lower Ground Rail)	0V System Common Ground Return
Red	Pin 5V (Power Header)	Red Bus Rail (+) (Lower Power Rail)	+5V Regulated DC Supply Rail

Table 2: Wire Connection Quick-Reference Matrix.

Phase 3: Breadboard Step-by-Step Wiring Sequence

Important Safety Rule

Keep the Arduino UNO **completely disconnected from USB power** while inserting components and jumper wires on the breadboard to prevent accidental shorts.

Step 2.1: Establish Main Power Rails

1. Take a **Red jumper wire**: Connect Arduino header pin **5V** to the **Red (+) power rail** on the breadboard.
2. Take a **Black jumper wire**: Connect Arduino header pin **GND** to the **Blue (–) ground rail** on the breadboard.

Step 2.2: Assemble the LDR Potential Divider (Sensor Subcircuit)

1. Insert the two legs of the **LDR (GL5528)** into **Row 10** and **Row 14** on the breadboard.
2. Connect a short jumper wire from **Row 10** (LDR Leg 1) to the **Red (+) 5V power rail**.
3. Insert one leg of the **10 k Ω resistor** (CFR-25JR-10K) into **Row 14** (shared with LDR Leg 2).
4. Insert the second leg of the **10 k Ω resistor** into the **Blue (–) Ground rail**.
5. Take a **Green jumper wire**: Connect from **Row 14** (the shared divider tap junction) to Arduino Analog Pin **A0**.

Step 2.3: Assemble the LED Luminaire (Actuator Subcircuit)

1. Insert the **5mm LED** (WP7113ID):
 - Insert the **Longer Lead (Anode +)** into **Row 20**.
 - Insert the **Shorter Lead (Cathode –)** into **Row 21**.
2. Insert the **220 Ω resistor** (CFR-25JR-220R) between **Row 20** (with LED Anode) and **Row 25**.
3. Take an **Orange jumper wire**: Connect from **Row 25** (the other side of the 220 Ω resistor) to Arduino Digital Pin **D9**.
4. Take a **Black jumper wire**: Connect from **Row 21** (LED Cathode) to the **Blue (–) Ground rail**.

Step 2.4: Pre-Power Visual Checklist

- ☐ LDR is connected between +5 V rail and Row 14.
- ☐ 10 k Ω resistor connects Row 14 to GND rail.
- ☐ Green wire connects Row 14 to Analog Pin A0.
- ☐ Orange wire connects Digital Pin D9 to the 220 Ω resistor.
- ☐ LED Anode (long leg) connects to the 220 Ω resistor; Cathode (short leg) connects to GND.

Phase 4: Firmware Flashing in Arduino IDE

Step 3.1: Connect Arduino to Workstation

Plug the USB Type-A to Type-B cable into the Arduino UNO and your computer's USB port. The green 'ON' LED on the Arduino board will illuminate steadily.

Step 3.2: Configure Board and Port in Arduino IDE

1. Open the **Arduino IDE**.
2. Go to: **Tools** → **Board** → **Arduino AVR Boards** → **Arduino Uno**.
3. Go to: **Tools** → **Port** and select the active COM port assigned to the board (e.g., COM3, COM4 — marked as "*Arduino Uno*").

Step 3.3: Load and Upload Sketch

1. Open the project sketch: `Automatic_Night_Light.ino`.
2. Click the **Verify** button (checkmark icon) to compile. Check that there are zero compiler errors.
3. Click the **Upload** button (right-arrow icon). The 'TX' and 'RX' yellow LEDs will flash rapidly for 2 seconds until the IDE displays: "**Done uploading**".

Phase 5: Live Calibration & Functional Testing

Step 4.1: Open Serial Monitor Telemetry

1. Click the magnifying glass icon at the top right corner of the IDE to launch the **Serial Monitor**.
2. Set the Baud rate dropdown at the bottom right corner to **9600 baud**.
3. Verify the live telemetry stream:

ADC: 839 | V: 4.10V | Luminaire: INACTIVE (OFF)

Step 4.2: Normal Room Light Baseline Test

Under standard room or lab lighting:

- Verify that the ADC value is between **600 and 850** ($V_{\text{out}} \approx 3.0\text{ V} - 4.2\text{ V}$).
- Confirm that the LED remains completely **OFF** (Daylight detected).

Step 4.3: Nightfall Simulation (Darkness Test)

Cover the LDR completely using your finger or an opaque object (e.g., black paper or notebook):

- Observe the Serial Monitor: the ADC value drops sharply below **400** ($V_{\text{out}} < 1.95\text{ V}$).
- Confirm: The LED turns **ON automatically and brightly!**
- The Serial Monitor reports: Luminaire: **ACTIVE (ON)** [Nightfall Detected].

Step 4.4: Hysteresis Deadband Verification (Anti-Flicker Demo)

Gradually pull your finger away from the LDR:

- Notice that when the ADC value climbs through the twilight transition region between **400 and 500**, the LED **remains solidly ON**. It does **NOT** flicker!
- Only when the light level rises fully above **500** ($V_{out} \geq 2.44\text{ V}$) does the LED turn OFF.
- This proves that the **100-count Schmitt trigger deadband** successfully prevents flickering at dusk!

Phase 6: Troubleshooting Reference Table

Observed Issue	Probable Cause	Step-by-Step Fix
LED never turns ON, even when LDR is covered.	1. LED inserted backwards. 2. Loose jumper wire in D9. 3. Burned-out LED.	Reverse LED leads (long lead connects to D9 via $220\ \Omega$). Verify wire is in Pin D9. Swap with another LED.
LED stays ON all the time, even in daylight.	1. $10\text{ k}\Omega$ resistor loose. 2. LDR not receiving $+5\text{ V}$. 3. Pin A0 floating.	Ensure $10\text{ k}\Omega$ resistor connects from node A0 to GND rail. Confirm LDR high side is in $+5\text{ V}$.
ADC readings jump wildly or erratically.	1. 50/60 Hz AC line noise. 2. Loose breadboard ties.	Firmware already includes a 10-sample moving average filter. Press wires firmly into breadboard tie holes.
Serial Monitor displays gibberish / strange characters.	Baud rate mismatch.	Change the baud rate dropdown at the bottom right of the Serial Monitor to 9600 baud .

Table 3: Common Fault Identification & Corrective Actions Matrix.

Laboratory Performance Evaluation & Sign-Off

Student Name: _____ **Roll Number:** _____ **Date:** _____

Circuit Assembly (4M): _____ **Firmware & Calibration (4M):** _____ **Viva-Voce (2M):** _____

Faculty Evaluator Signature: _____ **Total Marks:** / 10